

BUIDL CTC 2026 FALL · DEFI · RWA

ProofLine

Any app can read the credit file. Only verified proofs can change it.

Ethereum contains evidence of economic behaviour. Attestcoin lets Creditcoin verify that evidence. ProofLine turns verified events into persistent credit state that separate applications can consume.

Live on Ethereum Sepolia & Creditcoin CC3 · 124 tests · 8 contracts source-verified · every material claim reproducible from the repo

THE PROBLEM

Economic behaviour is provable. Creditworthiness is not portable.

A business earns on Ethereum. That history is public, permanent and cryptographically real - but it is not portable as independently verifiable credit state on another chain.

Bridging it means trusting an operator. Re-stating it means trusting a database. Both replace the one property that made the history worth having.

The question is not how to copy the history. It is how to make the credit decision depend on proof.

Remove the proof layer from ProofLine and the credit file cannot change at all

Attestcoin proves inclusion. It does not prove truth.

A proof establishes that a log was included in a transaction. ProofLine then verifies successful execution, event semantics and source authorisation. None of that establishes whether the log's claim is *true*.

So the source contract has to make a false claim unreachable. `InvoicePaid` cannot exist unless the measured ERC-20 balance delta equals the invoice amount. `InvoiceLate` cannot exist before the due date.

Attestcoin then proves the contract reached a state its own invariants made impossible to reach falsely.

SIX VERIFICATION GATES

- 1 replay proof-derived identity
- 2 verifyAndEmit precompile
- 3 transaction type
- 4 `receiptStatus = 1` protocol MUST
- 5 matching logs found
- 6 emitter authorised

Gate 6 must come last: the emitter address does not exist until the receipt is decoded, which requires the proof to have verified.

HOW IT WORKS

Ethereum Sepolia

Receivable.sol

real mUSD moves, invariants enforced on-chain

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Attestcoin

Merkle inclusion + continuity proof

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ASCReceiver.sol

six gates, proof-derived replay identity

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CreditFile.sol

← the primitive

append-only, proof-backed credit state

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├ UnderwritingLib

→ **Treasury.sol**

working capital

├ tier

→ **CreditAccess.sol**

security deposit

└ raw file

→ **NetTermsDesk.sol**

supplier terms · added ten days later

 One writer. CreditFile changes only when a proof verifies, in the same transaction

Five settlements changed the terms

SETTLEMENT	COUNTERPARTY	AMOUNT	TIER	ADVANCE	APR	DEPOSIT
#1	buyer A	\$12,000	STANDARD	60%	16%	75%
#2	buyer B	\$8,000	STANDARD	60%	16%	75%
#3	buyer C	\$10,000	GOOD	70%	14%	40%
#4	buyer A	\$7,000	GOOD	70%	14%	40%
#5	buyer B	\$9,000	TRUSTED	80%	12%	0%

Every row is a Sepolia transaction moving mUSD between three registered counterparties, all controlled by the builder, then verified through Attestcoin. No credit state was written directly: every tier change came from a verified settlement. **Capacity rose from 7,200 to 9,600 while the largest settlement stayed at 12,000** - every increase came from a rate earned by history, never from a bigger invoice.

THEN WE ACTUALLY LENT

Earned credit, used

STATE	TIER	CAPACITY	APPROVED LINE	AVAILABLE	DEBT
Before receivable	TRUSTED	9,600	9,600	0	0
Receivable outstanding	TRUSTED	9,600	9,600	9,600	0
Borrowed 6,300	TRUSTED	9,600	9,600	3,300	6,300
Repaid 3,150	TRUSTED	9,600	9,600	6,449.999	3,150.001

The first row is the architecture in one line. Earned standing of 9,600 with nothing currently drawable - a state that cannot be expressed at all if the three numbers are collapsed into one "credit limit".

Verified history changes earned capacity. Receivables and debt change availability. The .001 is accrued interest: 3,150.000719 on-chain.

Each row at its transaction's block • borrow 0xd3174e51 • repay 0x3cf490de • debt accrues at 12% APR after

THE DISTINCTION

Why this is infrastructure, not another lending app

REUSABLE

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Separate consumer contracts over one credit file. The third, NetTermsDesk, was deployed ten days later by a fresh address: it imports nothing, needed no permission, and CreditFile did not change.

LOAD-BEARING

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One writer. Delete Attestcoin and the credit file is immutable - not degraded, not stale. The financial state transition becomes impossible.

STATEFUL

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Credit is persistent state, not a one-shot reaction to a proof. History compounds; delinquency tightens terms immediately; a verified default freezes permanently.

One verified settlement moved Treasury from 70%/14% to 80%/12% and CreditAccess's deposit requirement from 40% to 0%, through a credit file that imports neither application. **The invoice contract is our source of evidence, not the product.**

Done, not claimed: NetTermsDesk 0x307C0Ecf · decision 0xde9b8fe9, net 60 · CreditFile code hash and 11 events unchanged

EVIDENCE & LIMITATIONS

MEASURED, NOT ASSUMED

- **7.96-8.7 min** end-to-end attestation
- **321,498 gas** mean production ingest, 8 samples
- **244-669 ms** proof construction once attested
- **Permissionless relay works** - the worker is convenience, not trust
- A **real Ethereum mainnet** transaction passed the same six gates

FOUND, FIXED, RECORDED

- Replay key was **caller-spoofable**
- Accounting silently **aliased its input**
- A gate-ordering test **passed while proving nothing**
- Unchecked payer allowed **manufactured counterparties**
- One of our claims was an inference - **disproved, corrected in place**

STATED PLAINLY

- **The demonstration history was seeded by the builder.** ProofLine verifies the events; it does not establish that counterparties are economically independent.
- **Repayment is unsecured.** Writability is in audit, so enforcement is credit-state based rather than collateral seizure. Deliberate, and Creditcoin's thesis.
- **Tier thresholds are our policy**, not a Creditcoin standard. The reusable part is the shape: verified history → deterministic function → terms.

WHY CREDITCOIN WANTS THIS

CreditFile gives Creditcoin applications a common, verifiable credit primitive instead of every application building its own cross-chain evidence pipeline.

verified economic activity → richer CreditFile
→ more applications → more Creditcoin demand

Every new source of verified history strengthens the same credit layer. Every new application can consume it.